

Auntu Gain

Embedded Systems & Robotics Engineer · 5+ years in Korea

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F-5 permanent resident (no work restrictions) · English: IELTS Academic 7.0 · Korean: TOPIK Level 5 · Engineer
Information Processing (정보처리기사)

SUMMARY

Embedded and robotics software engineer who builds software connecting sensors, perception and real-time control to physical systems. Experience spans MCU firmware on RTOS, vehicle and industrial communication (CAN/LIN, RS485, BLE, MAVLink), Linux/ROS2 sensor integration, computer vision on edge devices, and Qt/QML HMI. Currently developing production Hyundai commercial-vehicle instrument-cluster software.

SKILLS

Core	C, C++, Python · RTOS/FreeRTOS MCU firmware, STM32, BLE firmware · CAN, LIN, RS485, BLE, serial, MAVLink · UWB, LiDAR, ultrasonic and camera integration, BLDC and steering control · Linux, Raspberry Pi, Qt/QML · simulation, on-vehicle and field verification, interrupt/memory/watchdog debugging
Working experience	ROS2 (USV sensor integration), NVIDIA Jetson Nano (model deployment), PX4/Pixhawk, ArduPilot, QGroundControl · TensorFlow detection and pose estimation, OpenCV, CNN classification, anomaly detection (Isolation Forest, autoencoders) · Yocto image builds, Linux device drivers · MySQL, Google Cloud SQL, Git

EXPERIENCE

Associate Research Engineer · GT System Jul 2024 – Present · Yongin, Gyeonggi

Production automotive embedded software for Hyundai commercial-vehicle instrument clusters | C++, Qt/QML, CAN/LIN

- Implement cluster HMI functionality in C++ and Qt/QML: telltales, indicators, warnings and vehicle-status displays.
- Integrate CAN/LIN-derived sensor and ECU signals with application logic and display behavior.
- Translate functional requirements into software behavior; investigate issues across signal handling, application logic and HMI.
- Perform simulation-based testing and on-vehicle verification, reproduce defects and implement fixes; support unit testing, coverage review and integration verification.
- Coordinate development and verification tasks within an international engineering team.

Software Engineer · Approtec Inc. Jun 2023 – Jun 2024 · Seongnam, Gyeonggi

Robotic transport, automated water control and battery monitoring | UWB, BLDC, RS485, CAN, Qt, Raspberry Pi

- Agricultural follow-me vehicle: UWB and computer-vision target identification, following-distance control and obstacle avoidance; BLDC traction-motor and steering-position control software.
- Vanadium flow-battery monitoring: CAN and RS485 integration for flow, temperature, pressure and charge/discharge status; automatic inverter control and protective logic, enabling charge/discharge tests without continuous supervision.
- Water intake and recharge control: Raspberry Pi water-level/flow sensing over RS485, automatic pump control, Qt dashboard with role-based access, MySQL and Google Cloud SQL storage.
- Maintained deployed smart-shelter kiosk and IoT software (Gangnam-gu).

Software Engineer · Fluton Apr 2021 – Jun 2023 · Gyeonggi

Autonomous surface vehicles and connected-device firmware | C/C++, Python, RTOS, ROS2, Jetson, PX4

- Lifeguard USV: TensorFlow object detection and pose estimation on NVIDIA Jetson Nano to detect people in the water and trigger alerts during repeatable survey missions (related algorithm filed as a company patent application).
- Integrated LiDAR, ultrasonic and RealSense camera inputs with ROS2 to improve obstacle avoidance; connected perception to vehicle control via QGroundControl and custom MAVLink commands.
- Water-treatment skimmer drone (with K-water): Pixhawk/PX4 and RTK GPS for repeatable cleaning missions; AprilTag-based automatic docking and charging.
- Shared-bicycle firmware (contract): C firmware on an RTOS for locking, BLE, GPS, sensor monitoring and modem connectivity; BLE-to-MCU protocol; reduced heap use in GPS handling and resolved watchdog resets during MCU

sleep.

SELECTED INDEPENDENT WORK

- **Smart water-meter edge-AI module** (contract, ongoing): Isolation Forest and autoencoder leak/anomaly detection, lightweight models for STM32-based low-power hardware. The related technology is published as a patent under the client's name; now developing the first product's edge-AI module.
- **Robotic pill-sorting vision** (contract): CNN classification from high-resolution images integrated with robot control for real-time sorting, plus a retraining workflow for new pill classes.
- **Wingman AI co-driver** (personal): Android app for Hyundai Pleos Connect using Vehicle, NaviHelper and Gleo (STT/TTS/LLM) SDKs.

EDUCATION & OTHER

Education	B.S. Computer Science and Engineering, Jeonbuk National University (2017–2021) · Korean Government Scholarship (GKS) recipient
Research / intern	H&Y intern (2021): ArduPilot, QGroundControl, ChibiOS customization · JBNU Data Mining Lab undergraduate researcher (2018–2019)
Languages	English (IELTS 7.0) · Korean (TOPIK 5) · Bengali (native)
License	Korean driver's license, Class 1 Ordinary